

Main-Idea Extraction & Text Visualization

Related concerns can appear in different wording: “My parcel has not arrived” and “Where is my delivery?” You will discover keyphrases and topics, select traceable source excerpts, and visualize recurring ideas without confusing them with sentiment.

Objective Prepare a 100,000-row TWCS sample for main-idea extraction, reviewed topic discovery, extractive summaries and text visualization.

Specific technical objectives

- Select exactly 100,000 source rows using a deterministic SHA-256 rule and preserve identifiers, raw text, direction and time in an audit registry.
- Define distinct normalized-text representatives within inbound posts and document excluded rows, repeated content and unassigned text.
- Extract document frequencies and TF-IDF keyphrases with a sparse, bounded unigram/bigram vocabulary.
- Fit a fixed six-topic nonnegative matrix factorization, review topic labels and select representative source posts as extractive summaries.
- Create frequency, co-occurrence, topic-share and temporal visualizations with explicit denominators, source evidence and reproducible exports.

Expected learning outcomes

- Distinguish a keyphrase, a latent topic, a sentiment judgment and an extractive summary of social text.
- Calculate document frequency, TF-IDF, cosine similarity and a topic share on small examples.
- Explain why topic weights and human topic labels are exploratory descriptions rather than ground-truth categories or calibrated probabilities.
- Evaluate main-idea readability and source faithfulness through a fixed, documented human review protocol.
- Communicate what a sampled social-media corpus can support without generalizing to all customers or inventing sentiment labels.

Important Guardrails: TWCS inbound/outbound is message direction, not sentiment or intent. The corpus supplies no gold sentiment labels for this task. Customer-support posts are not a probability sample of public opinion. Preserve raw text, redact personal information in displays and keep source IDs for review. Complete this assignment independently; it requires no airline model or output from Lab 09A.

Run the demonstration

The TWCS portion is independent of airline predictions. Synthetic data cannot satisfy this assignment. Use source [B1] if needed; record its URL, acquisition date and file hash. Companion outputs use `_week09_build/text_nlp_outputs`; student artifacts use `Lab09B_outputs`.

Validate unique string `tweet_id` values while streaming the source. Hash the UTF-8 string “961731:” followed by each `tweet_id` using SHA-256. Sort ascending by digest, breaking a digest tie by `tweet_id`, and select exactly the first 100,000 distinct messages. Save the

selected IDs and digest. Conflicting duplicate IDs stop release; do not silently drop them to reach the quota. Chunk iteration is permitted for bounded ingestion, but repeated dataframe row slicing is not.

This is a fixed hash-based message sample, not a user sample. Filter nonempty inbound text after sampling and retain one earliest representative per exact normalized text. Preserve all 100,000 registry rows and report duplicate exclusions. Do not refill the sample. Topic results describe these text representatives, not message-weighted prevalence among all customers.

Dataset schema and ingestion contract

TWCS contains customer and brand messages with text, direction and timestamps. The five fields below define this assignment. Parent/reply identifiers may remain in the source but are not required for topic discovery. Parsing time supports descriptive text trends; a timestamp does not reveal the message's sentiment. [B1]

Field	Raw type	Canonical meaning	Intentional defect
tweet_id	ID string	Unique source message	Add whitespace to one stress-copy ID; detect any collision after trimming and never use floating-point identifiers.
text	string	Original message language	Add a mention, URL and empty string to stress-copy rows; verify normalization and preserve originals.
author_id	string	Source account key	Blank one stress-copy value. Keep identity unknown; do not merge every unknown author into one person.

Field	Raw type	Canonical meaning	Intentional defect
inbound	Boolean text	True: customer message; False: brand reply	Mix True/False casing and insert one unknown token. Use an explicit mapping; unknown is not False.
created_at	time string	Message creation instant	Insert an invalid stress-copy timestamp. Parse UTC explicitly and retain failures with raw values.

Injected defects are exercises, not source claims. Add `text_clean`, `created_at_utc`, `is_representative`, `analysis_eligible`, `exclusion_reason`, `feature_nonzero` and `topic_id`. For exact-text groups choose earliest valid UTC time, breaking ties by `tweet_id`; if all times are invalid, choose the smallest ID. Such representatives enter topic analysis but not time plots. Missing text never becomes a fabricated statement.

Problem 1 Corpus preparation and keyphrase extraction

A newspaper index lists phrases, while its editor groups related stories. Begin with the index. A frequent phrase may describe a concern, a brand or a copied template; frequency alone does not determine meaning.

Normalize Unicode/whitespace, replace mentions/URLs with markers, retain hashtag words and document contraction/emoji handling. Preserve negation in raw/clean text and source excerpts. Topic extraction uses a separate, declared stop list; removing negation there sacrifices polarity and must never define sentiment features or quotations. Show retained tokens and acknowledge English-tokenizer limits.

Use the installed scikit-learn `ENGLISH_STOP_WORDS` plus `usertoken`, `urltoken`, `hi`, `hello`, `thanks`, `thank`, `please`, `dm`, `does`, `did`, `just`, `help`, `need`, `got`, `ive`, `im`, `dont`, `hey`, `yes`, `ok`, `okay`,

really, still, like, want and know. Save the resolved list. Compare the initial smaller-list demonstration with this documented descriptive refinement; retain both results. Filtered-token bigrams may join words separated in the source, so call them candidate descriptors rather than guaranteed verbatim phrases.

Count a term once per document for document frequency, even if it appears repeatedly. Let B_{it} be one when document i contains term t and zero otherwise. Corpus token frequency and document frequency answer different questions.

$$df(t) = \sum_{i=1}^n B_{it}$$

For a term occurring c times in a post, use sublinear TF when c is positive and zero otherwise. Fit smoothed IDF on the declared analysis corpus, then L2-normalize each nonzero row. [B2]

$$TF^*(t, d) = 1 + \ln(c)$$
$$IDF(t, D) = \ln \left(\frac{1 + n}{1 + df(t)} \right) + 1$$

In “late parcel”, “late late delivery” and “thanks”, the unigram “late” has token count 3 but document frequency 2. Its IDF is $\ln(4/3)+1 \approx 1.288$; its sublinear TF in the second post is $1+\ln 2 \approx 1.693$. The bigram “late delivery” appears in one document. State whether each chart shows occurrences, documents or TF-IDF weight.

1.1 Implement the hash contract and save exactly 100,000 IDs. Audit direction, raw/clean emptiness, duplicate text, author counts and timestamps. Report a denominator flow from sampled rows to nonempty inbound posts, exact-text representatives, nonzero-feature rows and valid-time representatives. Record duplicate exclusions and explain why deduplication changes the population. Show stress-copy validation failures and the correction log.

1.2 Provide six raw/clean examples covering a hashtag, mention, URL, negation, emoji and mixed or non-English language if present. Reproduce the arithmetic above. Set `ngram_range` to (1,2), `max_features` to 5,000, `min_df` to 5 and `max_df` to 0.8, with sublinear TF, smoothed IDF and L2 normalization. Record the stop list and feature names. Assert nonnegative finite sparse values and flag zero vectors rather than silently deleting them.

1.3 Rank keyphrases by mean TF-IDF across eligible inbound text representatives and show the top 20 alongside document frequency. Select up to five highest-weight phrases for five fixed-seed examples. Use deterministic lexical tie-breaking. Compare useful phrases with vague fragments or templated language. A high TF-IDF weight marks distinctiveness within this representation, not verified importance, sentiment or a complete sentence summary.

The denominator flow, raw/clean examples, token-length distribution, document-frequency bars and TF-IDF phrase bars. This assignment fits all eligible text representatives for descriptive discovery; no held-out generalization score is claimed. A future predictive evaluation must fit vocabulary, IDF and topics on training data only.

Problem 2 Topic discovery and representative main ideas

Sorting a shelf into six provisional folders can reveal recurring subjects, but the folder names remain an interpretation. Nonnegative matrix factorization (NMF) approximates a nonnegative document-term matrix X with document-topic weights W and topic-term weights H . It groups patterns of co-occurrence; it does not read a customer's private intent.

[B3]

$$X \approx WH$$

For n documents, v terms and the fixed rank $K = 6$, X has shape n by v , W has shape n by 6 and H has shape 6 by v . Fit nonnegative factors using the Frobenius reconstruction

objective below. This simplified objective states the chosen unregularized teaching model, not every possible NMF variant.

$$\min_{W, H \geq 0} \frac{1}{2} \|X - WH\|_F^2$$

Topic factors have an arbitrary scale: multiplying a W column and dividing its H row by the same positive value preserves WH . Normalize each nonzero H row to unit L2 norm and multiply its W column by the original norm. Call these factors H' and W' ; reconstruction must remain unchanged.

$$q_{ik} = \frac{W'_{ik}}{\sum_{r=1}^6 W'_{ir}}$$

Use the largest q as a descriptive dominant topic, breaking ties by topic ID. These normalized weights are not calibrated probabilities or confidence. A zero feature/weight row is unassigned. Arithmetic bridge: weights (2,1,1,0,0,0) give (0.50,0.25,0.25,0,0,0). The first topic is largest, but the post still contains a mixture.

2.1 Fit exactly six topics with `random_state 961731`, `init nndsvda`, `max_iter 300` and `tol 0.0001`. Record convergence, reconstruction error, vocabulary size, nonzero count and runtime. Show each topic's ten largest term weights and assigned representative count. Normalize factor scales and verify reconstruction invariance on a bounded slice. Do not vary rank or seed until attractive labels appear; fixed rank is a teaching choice, not proof of six natural categories.

2.2 Draw 20 fixed-seed assigned posts per topic, or all if fewer. Retain source IDs and actual review counts. Propose a short label, then record whether each post fits that label, is mixed/ambiguous or is a poor match. Describe disagreements and revise the human label transparently; do not erase difficult examples or refit to hide them. Report readable-label

and source-fit counts with denominators, not an invented supervised accuracy. If two reviewers are available, record independent judgments before discussing differences.

2.3 Within each assigned topic, rank posts by cosine similarity to its normalized topic vector and select three distinct texts, breaking ties by tweet_id. Keep IDs, scores and redacted raw/clean wording with negation preserved. A summary card contains the reviewed label, candidate descriptors and source excerpts. Call it an extractive summary; never reconstruct quotes from filtered tokens or add unsupported events, entities or opinions. Explain redactions.

$$\cos(x, h) = \frac{x^\top h}{\|x\|_2 \|h\|_2}$$

For $x = (1,0)$ and $h = (1,1)$, cosine similarity is $1/\sqrt{2} \approx 0.707$. It measures direction in a feature space, not factual truth. The most representative posts can omit rare concerns; retain one poor-fit or mixed example alongside each topic's interpretation when observed. Zero-norm vectors have undefined cosine and remain outside representative ranking.

Problem 3 Text visualization and evidence-based interpretation

A map needs both place names and a scale. A text dashboard likewise needs interpretable labels and denominators. A large topic bar may reflect more sampled messages, while a larger share can occur even when its count falls. Display counts and proportions together before telling a trend story.

Use a bounded document-term presence matrix B for the 20 most frequent eligible unigrams. Pairwise document co-occurrence counts follow the matrix product below; the diagonal gives document frequency and is hidden in the network/heatmap display.

$$C = B^\top B$$

An edge says two words occur in the same post; it does not establish a grammatical relationship, sentiment or causation. Keep the top 40 positive off-diagonal edges with deterministic ties, or show a 20-by-20 heatmap. Document the display threshold instead of choosing a dense network that cannot be read.

For time bin b , n_b includes eligible inbound text representatives with valid timestamps, including unassigned rows. The numerator n_{kb} counts each representative once in its dominant topic. Show unassigned separately; this is representative-text share, not all-message prevalence.

$$s_{kb} = \frac{n_{kb}}{n_b}$$

30 of 100 posts gives 30%; 45 of 200 gives 22.5%. The count rises while the share falls. Empty bins have undefined shares, not zero. Convert times to UTC before forming a declared daily or weekly binning rule and identify partial collection periods.

3.1 Produce a labeled co-occurrence heatmap or bounded network, a six-topic top-term heatmap and a topic count/share chart including unassigned. Use readable text, restrained consistent colors and informative units. Avoid interpreting a word cloud or projection's arbitrary coordinates as validated semantic distances. Save the underlying plotted tables so every bar, cell and edge can be recalculated.

3.2 Show change over time Plot representative counts and topic proportions in predeclared bins, with n_b visible. Keep vocabulary, NMF factors and labels fixed across bins. Report invalid-time exclusions and small denominators; sampling or collection changes can resemble topic change. Earliest-representative selection also affects timing. Do not connect independently relabeled daily topics as if they were one trend.

3.3 Deliver a one-page findings board with three observations, their figures, reviewed topic labels and traceable excerpts. Use the Problem 2 audit to discuss readability, coverage,

mixture and source faithfulness. Distinguish topic prevalence from sentiment or customer satisfaction. Specify one follow-up annotation question and one failure case. Optional embedding, TextRank or language-model comparisons must retain source evidence, a bounded computation budget and a declared evaluation protocol; fluent generated prose is not proof of faithfulness.

Stream hashing and bounded chunk ingestion, then use sorting, vectorized strings, groupby and sparse products. Sorting is $O(N \log N)$; with fixed vocabulary, rank and iteration budget, NMF uses bounded sparse operations. Never build a dense n -by- n similarity matrix or use `iterrows()`. Assert 100,000 unique registry IDs, admissible direction values, finite nonnegative factors, stable dimensions and conserved rows. Topic counts plus unassigned must equal eligible representatives; proportions sum to one only in nonempty bins. Findings are descriptive. Optional daily regressions require Newey–West errors with documented automatic bandwidth and lag sensitivity; repeated-user inference must respect author clusters.

Deliverables and submission protocol

- An executed notebook covering Parts 1.1–3.3, small calculations, the six-topic model, human-review evidence, source-linked summary cards and all requested visualizations.
- `twcs_sample_registry.parquet` with 100,000 rows; `twcs_sample_ids.csv`, `keyphrases.csv`, `nmf_topic_terms.csv`, `topic_assignments.parquet`, `topic_summaries.csv`, `topic_review.csv` and `validation_report.json`.
- A data dictionary, sampling/source/defect manifest, saved vocabulary/model settings, figures with plotted tables and a two-page defense of sampling, topic interpretation and evidence limits.

Submit one ZIP named `961731_studentID_Lab09B.zip`. Include the notebook, source script, resolved requirements and source hash. Record the actual inbound, nonzero,

assigned and valid-time denominators. The notebook must run after a kernel restart without an output from Lab 09A. Preserve unfavorable or ambiguous findings and distinguish machine-generated weights from human-reviewed labels.

Evaluation rubric

Evaluation Rubric 100 points. Credit depends on reproducible text analysis, readable evidence and faithful interpretation, including a clear account of ambiguous or unassigned posts.

Criterion	Points	Full credit evidence
Diagnosis	25	10: deterministic sampling and ingestion audit; 10: normalization and denominator flow; 5: correct frequency, TF-IDF and similarity arithmetic.
Technical Execution	35	15: sparse keyphrase/six-topic workflow; 10: audited labels and traceable extractive summaries; 10: correct co-occurrence, share and temporal visualizations.
Methodological Defense	25	10: descriptive scope and sampling limits; 10: topic/sentiment distinction, human review and source faithfulness; 5: lecture and primary-source citations.
Automation	15	5: fresh-run reproducibility; 5: passing invariants and exports; 5: complete versioned submission with sample and source manifests.

Partial credit: award half of a component when its result exists but cannot be reproduced or justified. Award zero for missing evidence, hidden denominator changes, invented source statements or sentiment labels, fabricated reviews or an unresolved executable failure. A compelling topic name cannot replace a review of the posts assigned to it.

Sources and further reading

[B1] Thought Vector. Customer Support on Twitter. Preserve source attribution and applicable reuse terms; message direction is separate from sentiment. <https://www.kaggle.com/datasets/thoughtvector/customer-support-on-twitter>

[B2] Scikit-learn developers. Text feature extraction. The lab uses its documented sparse n-gram and TF-IDF representation.

[B3] Scikit-learn developers. Nonnegative matrix factorization and topic-extraction example. Topic naming and human review remain analyst responsibilities.

Local teaching sources: Week09 - NLP for Social Media Analysis.pptx; Week09 - NLP for Social Media Analysis - Notebook.ipynb and its .py companion; Week09 - Notebook Guide.md. Main ideas describe what is discussed; sentiment describes evaluative tone. Neither substitutes for the other.

[B4] Oyeboode et al., COVID-19 Pandemic: Identifying Key Issues using Social Media and Natural Language Processing (PDF pp. 3–4, 7; supplied arXiv:2008.10022v1).

[B5] Camacho-Collados et al., TweetNLP: Cutting-Edge Natural Language Processing for Social Media (PDF pp. 2–4, 6–8; supplied arXiv:2206.14774v3).

[B6] Buyanov, Valieva and Mazurina, psytechlab at CLPsych 2026: Utilising NLP methods and LLMs for Social Media Text Analysis (PDF pp. 3–5; supplied arXiv:2607.03003v1).

[B7] Jiang et al., Annotating the Tweebank Corpus on Named Entity Recognition and Building NLP Models for Social Media Analysis (PDF pp. 2–6; supplied arXiv:2201.07281v2).

[B8] Šćepanović et al., The Healthy States of America: Creating a Health Taxonomy with Social Media (PDF pp. 4–6; supplied arXiv:2103.01169v1).

Research support: Week09 - Research Reading Guide.md provides worked examples and source limitations for these connections. They support the existing tasks and introduce no extra deliverable or rubric component.